MIL-Y-51021
29 JUNE 1960
SUPERSEDING
(SEE SECTION 6)

MILITARY SPECIFICATION

YEAST AUTOLYSATE, DRIED, TECHNICAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of dried yeast autolysate for use as a constituent nutrient in biological media.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-D-723 — Drums, Fiber.

MILITARY

MIL-P-3803 — Plastic, Polyethylene, Molded and Extruded Shapes, Sheets and Tubing.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASSOCIATION OF OFFICIAL AGRICULTURAL CHEMISTS

Official Methods of Analysis, Eighth Edition, 1955.

(Application for copies should be addressed to the Association of Official Agricultural Chemists, P.O. Box 540, Benjamin Franklin Station, Washington 4, D. C.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Product.** The product shall be an unadulterated yeast autolysate made from a culture of *Saccharomyces cerevisiae* of good commercial quality.

3.1.1 *Autolysate.* The autolysate shall be a concentrate of the water soluble portion of the yeast cells. No chemicals other than an optional volatile plasmolyzer and the acids or bases required for preliminary pH adjustment shall be used for the production of the autolysate.

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3.1.2 *Volatile plasmolyzer.* The volatile plasmolyzer, when used, shall be ethyl acetate, chloroform or toluene.

3.2 **Odor.** The autolysate shall have a characteristic but not putrescent odor, and shall evidence no odor of residual volatile plasmolyzer.

3.3 **Biological assay.**

3.3.1 *Nutritional value.* The product covered by this specification shall supply sufficient nutritional value to support adequate growth of various micro-organisms (see 6.2).

3.3.2 *Pre-award samples.* At no cost to the Government, pre-award sample (see 4.4.2.1) from each lot intended for shipment under the contract or order shall be forwarded to the destination indicated by the contracting officer. These samples will be tested by the Government to determine compliance with 3.3.1.

3.4 **Certification.** The supplier shall certify that:

(a) The autolysate conforms to the requirements specified in 3.1.

(b) The lot or lots submitted to the Government for final acceptance are the lot or lots from which the pre-award samples were selected and approved by the Government as complying with the nutritional value requirement (see 3.3.1).

(c) During the period between the taking of pre-award samples and the time of shipment, the lot or lots were not subjected to any known condition which may have adverse effects on the nutritional value.

3.5 **Chemical and physical properties.** The chemical and physical properties of the autolysate shall conform to the requirements shown in table I when tested as specified in the test paragraphs referenced therein.

TABLE I. *Chemical and Physical Properties*

Property	Requirements*	Test paragraph
Total solids**.....	93.0% (min)	4.5.2
Total nitrogen**.....	10.0% (min)	4.5.3
Amino nitrogen**...	Not less than 35% of the total nitrogen content	4.5.4
Ash**.....	13.0% (max)	4.5.5
Sodium chloride**...	1.0% (max)	4.5.6
pH (in solution).....	5.0 to 6.2	4.5.7

* Percentage requirements are by weight.
** On an "as is" basis.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Objective evidence.** The supplier shall present objective evidence as required by the Government representative that all materials are in accordance with 3.1 (see 6.3).

4.3 **Alternative inspection (including testing) procedures.** The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the inspection procedures of this specification will govern.

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4.4 Inspection provisions.

4.4.1 *Regular production lot.* A regular production lot shall consist of the autolysate produced in the final drying operation by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials providing the process is continuous. In the event the process is a batch operation, each batch (see 6.4) shall constitute a lot.

4.4.2 *Sampling.*

4.4.2.1 *Pre-award samples.* A 50 gram (g) representative sample shall be removed from each of five containers selected at random from each lot. In the event there are fewer than five containers a sample shall be taken from each container. Each sample shall be placed in a separate, clean, dry container labeled in accordance with Standard MIL–STD–129 to identify the sample with the lot represented.

4.4.2.2 *Acceptance samples.* A 100-g representative sample shall be removed from each of five containers, selected at random from each lot, for testing separately as specified in 4.5.2 to 4.5.7 inclusive. In the event there are fewer than five containers a sample shall be taken from each container. Each sample shall be placed in a separate clean, dry container labeled to identify the sample with the lot represented.

4.4.2.3 *Examination (levels A and B preparation for delivery).* Sampling of filled containers shall be conducted in accordance with Standard MIL–STD–105.

4.4.3 *Inspection.*

4.4.3.1 *Odor.* The sample autolysate (see 4.4.2.2) shall be inspected for compliance with 3.2.

4.4.3.2 *Tests.* The sample autolysate (see 4.4.2.2) shall be tested as specified in 4.5.

4.4.3.3 *Examination (levels A and B preparation for delivery).* The sample filled containers (see 4.4.2.3) shall be examined for levels A and B preparation for delivery (if applicable) in accordance with the classification of defects and Standard MIL–STD–105.

4.4.3.4 *Classification of defects.*

4.4.3.4.1 *Levels A and B preparation for delivery* (sect. 5).

Categories	*Defects*
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Bag incorrect
102	Contents of bag incorrect
103	Fiber drum incorrect

4.5 **Tests.** Tests shall be conducted as follows:

4.5.1 *Instructions.* Use analytical reagent grade chemicals and distilled water throughout the tests. Where applicable, conduct blank determinations in parallel with the tests using the same quantities of the reagents, and apply corrections when significant. Unless otherwise specified, make all weighings to an accuracy of 1 milligram (mg).

4.5.2 *Total solids.* Weigh a 1-g specimen of the sample in a previously tared ground-glass stoppered weighing bottle, spreading the specimen evenly on the bottom of the bottle as it is introduced. Dry the specimen to a constant weight in a vacuum oven at a temperature of 80° to 85° C and an absolute pressure of 130 ± 5 millimeters of mercury. Calculate the percentage of total solids using the formula:

$$\text{Percent total solids} = \frac{\text{weight of dried specimen}}{\text{initial weight of specimen}} \times 100$$

4.5.3 *Total nitrogen.* Weigh a 1-g specimen of the sample in a previously tared 50-milliliter (ml) volumetric flask. Fill the flask to the mark with water and thoroughly mix the

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contents. Pipette 10.0 ml of this dilution into a Kjeldahl flask. (Save the remainder for the test specified in 4.5.4.) Determine the total nitrogen in accordance with paragraph 2.23 of Official Methods of Analysis of the Association of Official Agricultural Chemists.

4.5.4 *Amino nitrogen.* Pipette 5.0 ml of the dilution prepared for the test specified in 4.5.3 into a Van Slyke apparatus. Determine the amino nitrogen in accordance with paragraphs 23.20 and 23.21 of Official Methods of Analysis of the Association of Official Agricultural Chemists. Calculate the percentage of amino nitrogen on the basis of the total nitrogen (see 4.5.3).

4.5.5 *Ash.* Determine the ash content of the sample in accordance with paragraphs 23.24 and 29.12 of Official Methods of Analysis of the Association of Official Agricultural Chemists.

4.5.6 *Sodium chloride.* Weigh a 10-g specimen of the sample in a previously tared porcelain or nickel crucible. Add 1 ml of a concentrated solution of sodium hydroxide to the specimen and mix with a clean glass stirring rod. Wipe the rod with a wet filter paper and place the paper in the crucible. Heat the crucible in an oven at a temperature of 102° ± 2° C. until the water is expelled, and then place it in a muffle furnace at 525° C. until a white ash is obtained. Remove the crucible from the furnace and allow it to cool to room temperature. Dissolve the contents of the crucible in hot water and filter the solution into a 200-ml volumetric flask. Rinse the crucible and the filter paper several times with hot water and add the rinse water to the filtrate. Acidify the filtrate with 2 ml of concentrated nitric acid and add 50.0 ml of O.1N silver nitrate. Mix the contents of the flask, allow to cool, dilute the mixture to the mark, and mix thoroughly. Pipette 100.0 ml of the contents of the flask into a 250-ml Erlenmeyer flask, add 0.3 g of solid ammonium ferrosulfate, and titrate the mixture with O.1N ammonium thiocyanate. Calculate the percentage of sodium chloride as follows:

$$\text{Percent sodium chloride} = \frac{11.69\ (AB - CD)}{W}$$

where:

A = ml silver nitrate divided by 2.

B = normality of silver nitrate.

C = ml ammonium thiocyanate.

D = normality of ammonium thiocyanate.

W = weight of specimen in g.

4.5.7 *pH (in solution).* Dissolve a 2.00 g specimen of the sample in 50 ml of recently boiled water in a 100-ml volumetric flask, dilute to the mark with additional carbon dioxide free water and thoroughly mix the solution. Determine the pH of the solution at 25° C., using a glass electrode — saturated calomel electrode — pH meter system (see 6.5) in accordance with the instructions on the meter.

4.6 Rejection criteria. If any sample of autolysate fails to meet any one of the requirements specified herein, the lot represented shall be rejected. However, the supplier, at no expense to the Government shall have the option of conducting a partial or complete analysis of the contents of each container in the rejected lot. He may then remove the containers of defective autolysate, and resubmit the remaining portion of the lot for acceptance testing. If any sample from the resubmitted lot fails to meet any one of the requirements specified herein, the lot shall be finally rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* Approximately 75 pounds of the autolysate shall be sealed in a bag made of 0.0015-inch nominal thickness polyethylene conforming to Specification MIL–P–3803. The bag shall be placed in a fiber drum complying with Specification PPP–D–723, type II, grade A, with metal heads and closing rings.

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5.1.2 *Level B.* Approximately 75 pounds of the autolysate shall be sealed in a bag made of 0.0015-inch nominal thickness polyethylene conforming to Specification MIL-P-3803. The bag shall be placed in a fiber drum complying with Specification PPP-D-723, type I, grade A.

5.1.3 *Level C.* Autolysate shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Level of packing required.

(c) That pre-award samples will be required.

6.2 **Biological assay.** Due to unknown factors affecting the growth of various microorganisms and the possible danger in handling these organisms, the Government will conduct all tests on the bid samples to establish the nutritional value of the product. It is estimated that the prospective supplier will be notified of the acceptance or rejection of the product on the basis of nutritional value within 30 days after receipt of the pre-award samples.

6.3 **Objective evidence.** Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 **Batch.** For the purpose of this specification, a batch is defined as that quantity of material produced from a single charge of the final drying vessel.

6.5 **pH meters and systems.** pH meters and systems complying with the requirements of ASTM Designation E70 have been found to be satisfactory for pH determinations.

6.6 **Supersession.** This specification supersedes Chemical Corps Purchase Description 198-2-2B, dated 29 September 1959.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Army—Chemical Corps
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Preparing activity:
Army—Chemical Corps

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